

Multi-Team Scaling Scenarios

- **Scenario 1 - 2 Teams (24 employees in Total)**

- Team A (12 primary + 2 cross = 14 total)
- Team B (12 primary + 2 cross = 14 total)
- Each team has 2 cross-members borrowed from the next team.

- **Scenario 2 - 4 Teams (48 employees in Total)**

- Team A (12 primary + 2 cross = 14 total)
- Team B (12 primary + 2 cross = 14 total)
- Team C (12 primary + 2 cross = 14 total)
- Team D (12 primary + 2 cross = 14 total)
- Each team has 2 cross-members borrowed from the next team.

- **Scenario 3 - 8 Teams (96 employees in Total)**

- Team A (12 primary + 2 cross = 14 total)
- Team B (12 primary + 2 cross = 14 total)
- Team C (12 primary + 2 cross = 14 total)
- Team D (12 primary + 2 cross = 14 total)
- Team E (12 primary + 2 cross = 14 total)
- Team F (12 primary + 2 cross = 14 total)
- Team G (12 primary + 2 cross = 14 total)
- Team H (12 primary + 2 cross = 14 total)
- Each team has ~2 cross-members borrowed from the next team.

- **Scenario 4 - 16 Teams (192 employees in Total)**

- Team A (12 primary + 2 cross = 14 total)
- Team B (12 primary + 2 cross = 14 total)
- Team C (12 primary + 2 cross = 14 total)
- Team D (12 primary + 2 cross = 14 total)
- Team E (12 primary + 2 cross = 14 total)
- Team F (12 primary + 2 cross = 14 total)
- Team G (12 primary + 2 cross = 14 total)
- Team H (12 primary + 2 cross = 14 total)
-
- Team P (12 primary + 2 cross = 14 total)
- Each team has ~2 cross-members borrowed from the next team (20% cross-membership).

4 different scenarios for vacation generation:

- **Scenario 1 — Sequential Rotation (Base method)**
 - Employees take 30 consecutive vacation days, distributed sequentially throughout the year.
 - For example the first employee starts a 30 day block on January 1st and the following employee begins after the previous one ends (In this case on January 31st). If one of the employees have their vacation at the end of the year it will continue at the beginning of the year.
 - This guarantees that vacations days are evenly distributed
- **Scenario 2 — Random Two-Block Distribution**
 - Employees are assigned vacations on two random blocks of 15 days across the calendar year
 - Guaranteed that these two blocks won't overlap with one another (to make sure the total of vacation days is always 30)
- **Scenario 3 — Weekend Focused Short Breaks**
 - Considering employees can work every year during any day, this scenario was created to mimic a more real human vacation scenario
 - Employees take multiple 4–6-day breaks, distributed throughout the year and biased toward weekends
 - Each employee's set of breaks totals **30 days**, spread over ~6–8 mini-vacations spaced about 3 to 5 weeks apart.
- **Scenario 4 — Summer-Focused Staggered Vacations**
 - All employees' vacations are restricted to the **summer period (June 1 – August 31)**.
 - Each employee takes **one continuous 30-day vacation**, with start dates evenly distributed within this 3-month window.
 - Used to represent seasonal vacation spikes

Minimum Templates Generation

To simulate the operational staffing requirements corresponding to each multi-team scenario, a **Minimums Template Generator** was developed.

Scaling Methodology

For each target scenario (4, 8, and 16 teams),

The script performs proportional scaling and controlled variation as follows:

1. Per-Team Scaling Factor

The number of required employees per team is scaled by the ratio of target to base headcount:

$$Scale\ Factor = \frac{Employees\ per\ Team\ (Target)\ (excluding\ shared)}{Employees\ per\ Team\ (Base)}$$

For example, if each team in the base scenario has 10 employees and the target scenario has 7 per team, all the daily minimums are multiplied by 0.7. After multiplying they are then transformed into integers (using banker's rounding [e.g 2.5→2, 3.5→4])

2. Controlled Random Variation (± 1)

To avoid the templates to be exactly the same per team, a small random ± 1 perturbation was applied to a small fraction of days ($\approx 4-5\%$). These perturbations are balanced meaning that the number of +1 and -1 changes are equal to guarantee that the scaling remains constant.

3. Ideal Staffing

$$Ideal(d, t, e) = Minimum(d, t, e) + 1$$

To guarantee that if one of the members of a team has to miss a work day due to an unexpected situation the minimums are still covered.

Total Complete Scenarios -> 4 Multi-Team Scaling * Scenarios 4 Vacation Scenarios =

16 Complete Scenarios

Results

ILP (Maximum time - 90 minutes)

Weights used - 100 minimums, 1 ideals

	Time Taken	Minimums not accomplished	Ideals not accomplished	% of minimum not filled	% of ideals not filled
2Teams_Vacation1	1 min 14 sec	103	488	2.82%	8.36%
2Teams_Vacation2	1 min 14 sec	103	488	2.82%	8.36%
2Teams_Vacation3	1 min 16 sec	103	488	2.82%	8.36%
2Teams_Vacation4	1 min 55 sec	103	674	2.82%	11.54%
4Teams_Vacation1	9 min 06 sec	215	976	2.94%	8.35%
4Teams_Vacation2	9 min 27 sec	215	976	2.94%	8.35%
4Teams_Vacation3	8 min 24 sec	215	976	2.94%	8.35%
4Teams_Vacation4	5 min 08 sec	216	1378	2.95%	11.79%
8Teams_Vacation1	45 min 00 sec	412	1952	2.82%	8.35%
8Teams_Vacation2	56 min 01 sec	412	1952	2.82%	8.35%
8Teams_Vacation3	48 min 45 sec	412	1952	2.82%	8.35%
8Teams_Vacation4	36 min 07 sec	492	2783	3.36%	11.91%
16Teams_Vacation1					
16Teams_Vacation2					
16Teams_Vacation3					
16Teams_Vacation4					

CSP (Maximum time - 90 minutes)

Weights used - 100 minimums, 1 ideals

	Time Taken	Minimums not accomplished	Ideals not accomplished	% of minimum not filled	% of ideals not filled
2Teams_Vacation1	4 min 09 sec	103	488	2.82%	8.36%
2Teams_Vacation2	5 min 44 sec	103	491	2.82%	8.41%
2Teams_Vacation3	2 min 43 sec	103	488	2.82%	8.36%
2Teams_Vacation4	13 min 13 sec	103	681	2.82%	11.66%
4Teams_Vacation1	90 min 03 sec	215	976	2.94%	8.36%
4Teams_Vacation2	90 min 03 sec	215	976	2.94%	8.36%
4Teams_Vacation3	78 min 04 sec	215	976	2.94%	8.36%
4Teams_Vacation4	90 min 04 sec	216	1383	2.95%	11.84%
8Teams_Vacation1	90 min 13 sec	412	1952	2.82%	8.36%
8Teams_Vacation2	90 min 13 sec	412	1952	2.82%	8.36%
8Teams_Vacation3	79 min 08 sec	412	1952	2.82%	8.36%
8Teams_Vacation4	90 min 13 sec	492	2784	3.36%	11.91%
16Teams_Vacation1					
16Teams_Vacation2					
16Teams_Vacation3					
16Teams_Vacation4					

Note: The 16-team instances were attempted for both ILP and CSP, but all runs exceeded **2 hours** of runtime and did not finish; therefore, results are not reported for those rows.

Greedy Randomized (Maximum time - 90 minutes)

	Time Taken	Minimums no accomplished	Ideals not accomplished	% of minimum not filled	% of ideals not filled
2Teams_Vacation1	0m 03s	111	620	2.86%	10.62%
2Teams_Vacation2	0m 03s	112	641	2.86%	10.98%
2Teams_Vacation3	0m 03s	110	611	2.86%	10.46%
2Teams_Vacation4	0m 06s	147	782	2.87%	13.39%
4Teams_Vacation1	0m 06s	231	1285	3.16%	11.00%
4Teams_Vacation2	0m 05s	238	1296	3.26%	11.10%
4Teams_Vacation3	0m 06s	235	1252	3.22%	10.72%
4Teams_Vacation4	0m 11s	304	1582	4.16%	13.54%
8Teams_Vacation1	0m 11s	445	2518	3.05%	10.78%
8Teams_Vacation2	0m 11s	444	2542	3.04%	10.88%
8Teams_Vacation3	0m 12s	446	2476	3.05%	10.60%
8Teams_Vacation4	0m 13s	679	3141	4.65%	13.45%
16Teams_Vacation1	0m 25s	867	4905	2.97%	10.50%
16Teams_Vacation2	0m 24s	881	5072	3.02%	10.86%
16Teams_Vacation3	0m 25s	880	4978	3.01%	10.65%
16Teams_Vacation4	0m 25s	1222	6259	4.18%	13.40%

Heuristic Solver (New) (Maximum time - 90 minutes)

	Time Taken	Minimums not accomplished	Ideals not accomplished	% of minimum not filled	% of ideals not filled
2Teams_Vacation1	0m 01s	103	678	2.82%	11.61%
2Teams_Vacation2	0m 01s	103	678	2.82%	11.61%
2Teams_Vacation3	0m 01s	103	678	2.82%	11.61%
2Teams_Vacation4	0m 01s	106	801	2.90%	13.72%
4Teams_Vacation1	0m 01s	215	1421	2.94%	13.54%
4Teams_Vacation2	0m 01s	215	1363	2.94%	10.72%
4Teams_Vacation3	0m 01s	235	1345	2.94%	11.10%
4Teams_Vacation4	0m 01s	304	1599	2.95%	11.00%
8Teams_Vacation1	0m 01s	412	2657	2.82%	11.37%
8Teams_Vacation2	0m 01s	412	2683	2.82%	11.49%
8Teams_Vacation3	0m 01s	412	2733	2.82%	11.70%
8Teams_Vacation4	0m 01s	556	3251	3.81%	13.92%
16Teams_Vacation1	0m 01s	818	5245	2.80%	11.23%
16Teams_Vacation2	0m 02s	818	5447	2.80%	11.66%
16Teams_Vacation3	0m 02s	818	5359	2.80%	11.47%
16Teams_Vacation4	0m 02s	935	6423	3.20%	13.75%

Heuristic Solver With 100 Repeats (Maximum time - 90 minutes)

	Time Taken	Minimums not accomplished	Ideals not accomplished	% of minimum not filled	% of ideals not filled
2Teams_Vacation1	0m 07s	103	641	3.53%	10.98%
2Teams_Vacation2	0m 06s	103	660	3.53%	11.30%
2Teams_Vacation3	0m 07s	103	654	3.53%	11.20%
2Teams_Vacation4	0m 07s	105	780	3.60%	13.36%
4Teams_Vacation1	0m 13s	215	1363	3.27%	11.67%
4Teams_Vacation2	0m 13s	215	1336	3.27%	11.44%
4Teams_Vacation3	0m 14s	215	1318	3.27%	11.28%
4Teams_Vacation4	0m 14s	230	1596	3.50%	13.66%
8Teams_Vacation1	0m 30s	412	2632	2.97%	11.27%
8Teams_Vacation2	0m 29s	412	2659	2.97%	11.38%
8Teams_Vacation3	0m 29s	412	2671	2.97%	11.43%
8Teams_Vacation4	0m 30s	548	3244	3.95%	13.89%
16Teams_Vacation1	1m 09s	818	5232	2.87%	11.20%
16Teams_Vacation2	1m 10s	818	5363	2.87%	11.48%
16Teams_Vacation3	1m 14s	818	5341	2.87%	11.43%
16Teams_Vacation4	1m 14s	922	6365	3.24%	13.62%

GA

Weights used - 100 minimums, 1 ideals

The 3-shift GA introduces an additional shift (night) and enforces a no-backward-shift constraint, meaning an employee cannot work a night shift followed by a morning shift the next day. All other settings match the 2-shift configuration. There are 10 independent runs per scenario, the population size is 200, NBTS crossover is used, demand-guided mutation is used, the tournament size is 7, early stopping with patience is used and the minimum delta is 1. Results represent the average over ten runs.

	Time Taken	Minimums not accomplished	Ideals not accomplished	% of minimums not filled	% of ideals filled	Avg. Generations to Stop
2Teams_Vacation1 2Teams_Vacation2 2Teams_Vacation3 2Teams_Vacation4	6 min 12 sec	103.0	629.9	2.82%	10.79%	650.7
4Teams_Vacation1 4Teams_Vacation2 4Teams_Vacation3 4Teams_Vacation4	10 min 59 sec	215.2	1485.9	2.95%	12.72%	528.0
8Teams_Vacation1 8Teams_Vacation2 8Teams_Vacation3 8Teams_Vacation4	28 min 32 sec	417.8	2969.8	2.86%	12.71%	604.9
16Teams_Vacation1 16Teams_Vacation2 16Teams_Vacation3 16Teams_Vacation4	76 min 58 sec	847.0	5854.1	2.90%	12.53%	644.5
32Teams_Vacation1 32Teams_Vacation2 32Teams_Vacation3 32Teams_Vacation4	273 min 08 sec	1729.7	11501.7	2.96%	12.31%	810.7

Vacation scenarios 2, 3 and 4 to be run on reference machine for publication.

GA + Local Search

The same post-processing local search that was applied in the two-shift case is used here, but adapted for three shifts. It alternates between a day swap operator and a cyclic exchange operator until convergence, targeting ideal coverage only. All constraint repairs (including no backward shift) are verified after each swap. The improvement over the raw GA output is greater in the three-shift case due to the greater number of ideal slots available for reassignment.

	Time Taken	Ideals not accomplished	% of ideals not filled	Total Time (GA + LS)	
2Teams_Vacation1 2Teams_Vacation2 2Teams_Vacation3 2Teams_Vacation4	0.47 sec	513.5	8.79%	6 min 13 sec	
4Teams_Vacation1 4Teams_Vacation2 4Teams_Vacation3 4Teams_Vacation4	2.47 sec	1127.6	9.65%	11 min 01 sec	
8Teams_Vacation1 8Teams_Vacation2 8Teams_Vacation3 8Teams_Vacation4	3.83 sec	2160.7	9.25%	28 min 36 sec	
16Teams_Vacation1 16Teams_Vacation2 16Teams_Vacation3 16Teams_Vacation4	5.12 sec	4193.8	8.98%	77 min 03 sec	
32Teams_Vacation1 32Teams_Vacation2 32Teams_Vacation3 32Teams_Vacation4	10.68 sec	8311.0	8.89%	273 min 19 sec	

Vacation scenarios 2, 3 and 4 to be run on reference machine for publication.

Discussion

All approaches keep the % of minimums not filled relatively low (generally around **2.8–3.4%** for ILP/CSP up to 8 teams, and below **5%** even for the heuristics). In contrast, the % of ideals not filled is substantially higher across all methods (typically **8–12%**, and reaching **13–14%** in the most difficult scenarios). This indicates that the system is operating close to the feasibility boundary: minimum coverage is mostly achievable, but adding the “+1” protection for robustness is frequently not possible without adding more employees.

Solver performance and scalability

A quality-runtime tradeoff can clearly be seen:

- **ILP** produces the best overall quality among the methods with recorded results, with minimum shortfalls staying near **2.82–2.95%** for Vacation Scenarios 1–3 (up to 8 teams), and ideal shortfalls near **8.35–8.36%**. Runtime increases steeply with scale, from about **1–2 minutes** (2 teams) to roughly **36–56 minutes** (8 teams). The 16-team ILP rows are present but contain no values in the results table.
- **CSP** achieves solution quality that is broadly comparable to ILP on the reported cases, but it exhibits much weaker scalability in runtime: for 4 and 8 teams, many runs reach approximately the **90-minute** cap. As with ILP, the 16-team CSP rows are shown without reported values.
- **Greedy Randomized** is extremely fast, taking only a few seconds, but it sacrifices quality: minimum shortfalls rise as high as **4.65%** (8 teams, Vacation 4), and ideal shortfalls rise to about **13.5%** in the most constrained cases.
- **Heuristic Solver (single run)** is also extremely fast (1–2 seconds) and generally improves on Greedy Randomized in minimum coverage (e.g., for 16 teams Vacation 1–3 it reports **2.80%** minimums not filled vs. about **3.0%** for Greedy). However, it still has noticeably worse ideal coverage than ILP/CSP, with ideal shortfalls commonly around **11–12%** and peaking near **13.9%**.
- **Heuristic Solver with 100 repeats** (multi-start) increases runtime but remains practical at all tested scales: about **6–7 seconds** (2 teams), **13–14 seconds** (4 teams), **29–30 seconds** (8 teams), and about **1:09–1:14 minutes** (16 teams). Compared to the single-run heuristic, repeats consistently reduce the number of **ideals not accomplished** (e.g., 2 teams Vacation 1: **678** → **641**; 8 teams Vacation 3: **2733** → **2671**; 16 teams Vacation 2: **5447** → **5363**). Minimum shortfalls are mostly unchanged in the easier cases and improve slightly in the most constrained settings (notably Vacation 4, e.g., 16 teams: **935** → **922** minimums not accomplished). Overall, repeats provide a modest but consistent quality boost for a small additional runtime cost.
- **GA** (results averaged over **10 runs**) matches ILP and CSP on minimum coverage (e.g., **103** minimums not filled for 2 teams across all methods), but results in a higher ideal shortfall of around **10.79%** for 2 teams, compared to **8.36%** for the ILP/CSP methods. The post-

processing local search consistently reduces this gap, bringing the 2-team ideal shortfall from **10.79% → 8.79%**, and showing proportionally larger improvements at 4 and 8 teams (e.g., 8 teams: **12.71% → 9.25%**). The main advantage of the GA is **scalability**: it successfully produces results for **16 and 32 teams** within a feasible runtime, scenarios where both ILP and CSP exceed the **90-minute** cap and report no solution.

Overall, the results suggest that **exact optimization (ILP)** is practical up to at least **8 teams** under the time cap and provides the strongest solution quality, while **CSP becomes time-limited earlier**. For larger scales (e.g., 16 teams), the heuristic methods (single-run and multi-start) are the only approaches with fully reported outcomes, indicating strong practical value for large instances when runtime is critical.

Overall, the results suggest that **exact optimization (ILP)** is practical up to at least **8 teams** under the time cap and provides the strongest solution quality, while **CSP becomes time-limited earlier**. For larger scales (e.g., 16 teams), the heuristics are the only methods with fully reported outcomes, indicating strong practical value for large instances when runtime is critical.

Impact of vacation scenario design

Vacation generation method has a consistent effect on difficulty. In nearly every algorithm, **Vacation Scenario 4 (Summer-focused staggered vacations)** is the hardest case:

- It increases the **ideal shortfalls** substantially (e.g., ILP ideals not filled rise from about **8.35%** in Scenarios 1–3 to **11.54–11.91%** in Scenario 4 across 2–8 teams).
- It is also the only scenario that clearly increases **minimum shortfalls** at larger scales (e.g., for 8 teams, minimums not filled increase to **3.36%** under ILP/CSP, versus **2.82%** for the other vacation scenarios).
- The heuristic and greedy methods show the same pattern, with their worst minimum and ideal percentages occurring under Vacation 4.

This is very intuitive since a lot of employees are having their vacations in a specific narrow seasonal window, will reduce staffing flexibility and make it harder not only to reach ideals but also to consistently meet minimums.